

Stability of Soybean Harvest Index¹

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ABSTRACT

Previous work had indicated that the harvest index within cultivars of soybeans [*Glycine max* (L.) Merr.] might be fairly stable. Such stability of harvest index would greatly simplify the conceptualization of the influence of seed growth processes on yield. In the studies reported here harvest index of field-grown soybean plants was measured in response to four types of environmental influence: interplant competition, extended photoperiod, timing of drought stress, and degrees of drought stress (line source irrigation). The experiments included both indeterminate and determinate cultivars, and included both a northern (Ithaca, N.Y.) and southern location (Gainesville, Fla). These experimental conditions resulted in considerable variability in plant sizes within a cultivar but the harvest index (ratio of seed mass to mass of mature plant without fallen leaves and petioles) was stable. Total masses of individual mature 'Chippewa 64' and 'Bragg' plants varied substantially in response to interplant competition, but harvest index was essentially constant (0.54 and 0.50 for Chippewa 64 and Bragg, respectively) across most of the range of plant sizes. Harvest index of smaller plants was less than for larger plants, but these smaller plants have little influence on total crop yield and harvest index. Extending the photoperiod to 14.5 h throughout the life cycle of six cultivars increasingly delayed ontogenic events as the distance from the photoperiod lamps decreased. However, no influence on harvest index was observed except at the position nearest the lamps where the standard errors of the harvest indices were increased. Harvest index did not respond to differences in the timing of severe drought stress. When rain was withheld from 'Wilkin' plants during vegetative growth, flowering, or seed setting, harvest index of individual plants was essentially 0.60. Only those plants stressed late in pod filling had lower harvest index and increased variability in harvest index. Seed yields (Chippewa 64, Wilkin, and 'Evans' for three planting dates) decreased substantially as degree of drought stress, induced by increasing distance from a line source irrigation system, increased yet, harvest index changed negligibly with distance from the line source. Thus, we conclude that harvest index is a conservative characteristic within each soybean cultivar.

Additional index words: *Glycine max* (L.) Merr., Drought stress, Interplant competition.

A RECENT simulation of leaf production, light interception, and conversion of solar radiation to plant mass (Sinclair and Spaeth, 1982) predicted total plant mass of a soybean (*Glycine max* L. Merr.) crop through a growing season. If the harvest index of a crop, the fraction which seeds contribute to total biomass (Donald and Hamblin, 1976) were known, its seed yield could also be predicted. A detailed analysis of component compensation within two soybean cultivars (Spaeth and Sinclair, 1984) identified a distinctive pattern in the relationships among total mass of mature individual plants, the mean mass per seed and seed number per plant. Graphs of mean mass per seed vs. seed number per plant formed hyperbolic iso-quants among individual plants of equal total mass, which suggested that harvest index was constant within a

cultivar. If harvest index were generally constant within cultivars it would be a useful tool in prediction of seed yield from total biomass.

Spaeth and Sinclair (1984) did not examine whether interplant competition, which induced differences in individual plant size, influenced harvest index. Wilcox (1974) measured harvest index of individual soybean plants and found that harvest index had low variance compared with the variance of absolute total masses (cv. 6.6 and 19.4%, respectively). Johnson and Major (1979) found that harvest index of soybeans was insensitive to planting date except in cases of killing frosts. These previous studies produced limited data and suggested the need for additional study of variation in the harvest index among individuals of a population and among treatments representative of common agronomic conditions.

In addition to prediction of seed yield, the variability of harvest index will allow assessment of the control of distribution of materials at the organizational level of the whole plant. A relatively large variability would suggest either: a) lack of genetic homogeneity in the population, b) lack of control of distribution of dry matter at the whole plant level, or c) high degree of sensitivity to microenvironmental characteristics which differ among individuals but which are not closely related to plant size. On the other hand, small variability of harvest index among individuals of a cultivar would imply a close coupling between processes of vegetative and reproductive development irrespective of plant size. Harvest index would then establish a fundamental similarity among individuals which appear to be quite different from one another.

This research was undertaken to characterize the variation of harvest index within soybean cultivars grown under various agronomic conditions. Data were collected on possible variations in harvest index from four experimental conditions inducing variation in growth among plants within cultivars: (1) a large number of individual plants were harvested from plots where interplant competition resulted in substantial variability in plant size, (2) plants were grown under extended photoperiods so that marked variability in plant ontogeny was induced, (3) drought stress was imposed on plants at different stages of plant development, and (4) the total quantity of water received by plants was varied to limit growth. In all conditions substantial variability in final plant size was found.

MATERIALS AND METHODS

All harvest index data were obtained by harvesting mature plants from which the senesced leaves had fallen. Schapaugh and Wilcox (1980) compared the harvest indices obtained by either including or excluding fallen leaves and petioles in a 2-year study using 24 soybean cultivars. They found a high correlation between the two methods for calculating harvest index in each of the 2 years ($r^2 = 0.93$ and 0.96). Consequently only the standing, mature plant was harvested by cutting the stem at the cotyledonary node. The shoots were dried at 35 °C, weighed, and then the seeds separated from

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the pods by hand and weighed. The harvest index was calculated as the ratio of seed mass to the total harvested shoot mass.

The hypothesis for these studies was that there was little or no variability in harvest index within a cultivar even though large variability may exist in plant and total seed mass. While statistical tests (e.g., Duncan Multiple Range Test) were used, generally these were of little use. In many cases no significant difference was shown, but this did not necessarily prove that no differences existed. Also in those cases where differences were statistically demonstrated, the magnitude of the differences in harvest index were generally small and the statistical conclusion was based on small variances about the means. Consequently the practical, agronomic significance of such statistical differences in harvest index were quite small. The data are presented as means and standard error of the means so the practical significance of variability in harvest index can be judged.

Responses to Interplant Competition. Soybean cultivars Chippewa 64 and Bragg were grown at Tailby Field, Varna, N.Y. and the Agronomy Research Farm, Gainesville, Fla., respectively, for study of variation in harvest index among individual plants. The soil at Tailby Field, N.Y. was a Hudson silt loam (fine, illitic mesic Glossaquic Hapludalf) which was fertilized with 7.2 kgN, 12.7 kgP, and 23.9 kgK/ha. Alachlor [2-chloro-2',6'-diethyl-N-(methoxymethyl) acetanilide] and metribuzen [4-amino-6-tert-butyl-3-(methylthio)-as-triazin-5(4H)-one] were applied preemergence at 2 and 0.4 kg/ha, respectively. Chippewa 64 seeds were inoculated with commercial *Rhizobia japonicum* inoculant before sowing on 14 May 1979. Seeds were sown at a rate of 26 seeds/m in rows 0.76 m apart and 6.5 m long. Seeds derived from a single parent plant harvested in 1978 (a single parent line) were sown individually at 0.36 m intervals in the rows of the bulk population and tagged at emergence. Plants were thinned to 20 plants/m before they reached Stage V-4 (Fehr and Caviness, 1977). Plants first flowered on 7 July. Seeds in pods at low nodes began dry matter accumulation on 20 July. Individual plants were harvested as they reached harvest maturity between 7 and 15 September. Seven of the single-parent line plants were harvested at maturity and 40 plants were sampled at random from the bulk population.

The soil at the Agronomy Research Farm, Florida was an Arredondo fine sand (hyperthermic, coated, Typic Quartz Psamment) which was fertilized with 0 kgN, 22 kgP, and 83 kgK/ha. Alachlor was applied preemergence at 2 kg/ha active ingredient (a.i.). Seeds of Bragg were sown on 24 June 1981 at a rate of 33 seeds/m in rows 0.96 m apart and 40 m long. Two-hundred and fifty mature plants were individually harvested on 28 October by sampling all plants in two 5 m sections of row.

Responses To Extended Photoperiod. This study was done at the Agronomy Research Farm, Florida during 1981. Tungsten-halogen lamps (1500 W) were mounted on poles at a height 2.5 m above the soil as described by Sinclair (1984). Four lamps spaced 6.4 m apart were arranged in a line at the east side of the plots. The lamps were controlled by a timer so the plants were subjected to a 14.5-h daylength throughout their life cycles. On 16 March, cultivars 'Chippewa 64', 'Williams', and 'Guelph' were seeded into rows perpendicular to the line of lamps. Eight rows of each cultivar were seeded. Five plants were harvested at 1, 4, 7, 10, 15, and 20 m from the lamps after all pods had browned at that position. Following the harvest of these three indeterminate cultivars, three additional cultivars ('Hardee', Bragg, and 'Biloxi') were seeded on 24 June. At maturity 0.5 m of row was harvested at each of the six positions from the lamps. The agronomic practices for all cultivars were the same as described in the previous paragraph.

Responses to Timing of Drought Stress. 'Wilkin' plants for the study of the influence of timing of drought stress, were grown at Tailby Field, Varna, N.Y. in 1979. Cultivation

methods were similar to those which were described previously for Chippewa 64 at this location. An additional four row by 10 m block was sown 3 weeks after the main sowing. Soils in the part of the field where Wilkin was grown were a combination of Red Hook gravelly silt loam [coarse-loamy, mixed, nonacid, mesic Aeric Haplaquepts], Dunkirk silt loam, (fine-silty, mixed, mesic Glossoboric Hapludalfs), and Braceville gravelly loam (coarse-loamy, mixed, mesic Typic Fragiochrepts). The timing of drought stress was controlled by excluding rainfall with shelters and adding water with irrigation (Vendeland et al., 1982). Two shelters measuring 11 × 8 m were used in this study and moved over the drought stress plots only during periods of rain. Drought stress was imposed at four different stages of crop growth to give four treatments: (1) vegetative, 14 May to 9 July; (2) flowering, 14 June to 20 July; (3) seed set, 9 July to 6 August; (4) seed fill, 23 July to harvest. An additional set of plants (treatment 0) received rainfall throughout the season, and irrigation when rainfall was insufficient. Another well-watered treatment (5) was established by sowing the seeds 3 weeks later than the other treatments. The number of plants harvested in treatments 0 through 5 were 13, 12, 6, 6, 13, and 10 plants, respectively.

Responses To Irrigation Rates. Three sowings of three cultivars (Chippewa 64, 'Evans', and Wilkin) were grown at Gainesville, Fla. to identify seed yield and harvest index response to different quantities of water. On 6 Mar. 1981, 0 kgN, 26.4 kgP, and 99.6 kgK/ha were broadcast and disked into the Arredondo fine sand soil. Rows ran East-West and were 0.91 m apart and 45 m long. Eighteen-row blocks (six continuous rows per cultivar) were sown on 16 March, 2 and 16 April. Population density was about 350 000 plants/ha. After each sowing alachlor (2 kg/ha a.i.) was applied as a preemergence herbicide. Blocks were irrigated uniformly from the time of sowing through 5 May. The blocks were then irrigated on 12, 15, 21, 22, and 23 May from a line source configuration (Hanks et al, 1976). Sprinklers on 1.5 m risers were spaced at 20 m intervals on an irrigation line positioned in the middle of the blocks perpendicular to the row direction. Twenty millimeters of water were applied at each irrigation beneath the line source. At the 12 May irrigation the Chippewa 64 plants in the first, second, and third sowings were at growth stages R6, R3, and V4 (Fehr and Caviness, 1977), respectively. Evans plants were at stages R5, R3, and V3, respectively. Wilkin plants were at stages R6, R3, and V3, respectively. On 27 May the nonuniform water distributions were terminated. Plots were harvested at maturity on 10 and 16 June and 4 July for the three sowing dates, respectively. Samples consisted of 1-m sections of row. Distances from the line source (12.9, 20.4, 27.0, and 33.0 m) were measured to sample centers. Three samples were taken from each combination of cultivar, sowing date, and distance.

RESULTS

Responses To Interplant Competition. Individual plants from populations grown in the field varied substantially in mature plant mass as anticipated, so that variability was not merely the result of a small number of outlying individuals. Thus, the microenvironmental conditions for plant growth differed substantially among individuals within a row.

The variation in harvest index associated with these differences in plant size was consistent. Harvest indices of Chippewa 64 and Bragg increased from low values for the smallest plants to constant values of 0.54 and 0.50, respectively, above a threshold in plant size (Fig. 1 and 2). Plants were distributed across a broad range of masses, yet most plants produced a seed mass in direct proportion to the vegetative mass irrespective

(Table 3). The lower harvest index and greater variation in treatment 4 was associated with the presence of green color in stems even after plants were completely dry. But even that variation was relatively small.

Responses To Irrigation Rates. Seed yield and harvest index differed in the response to irrigation rates. Data from the third sowing of Chippewa 64 (Fig. 3) illustrate the pattern of large yield reductions with distance from the line source observed at all sowing dates. In contrast, harvest index was essentially constant across all four irrigation rates (Fig. 3). Shoot masses, seed yields, and harvest indices of Chippewa 64, Evans, and Wilkin for all three sowing dates exhibited similar responses to variation in water supply. Total shoot mass decreased significantly across the gradient from most to least water in eight of nine cultivar and date combinations (Table 4). Seed yields reflected those decreases in biomass. Those furthest from the line source averaged 61% of the yields nearest the source. In fact, regressions of seed yield on shoot mass accounted for 98, 98, and 97% of the variation in Chippewa 64, Evans, and Wilkin yields, respectively, across all sowing dates and quantities of water.

DISCUSSION

Harvest index of a cultivar was a stable characteristic irrespective of substantial differences in environmental conditions during plant growth. Differences among individuals from the same plot included large differences in total plant mass, but only for the smallest plants was the harvest index reduced. An extended photoperiod that greatly influenced plant ontogeny had no influence on harvest index within a cultivar in virtually all cases. Harvest index of Wilkin was stable despite the imposition of severe drought stress at four stages of plant development. Seed yield decreased when plant growth was restricted by less water, but harvest

index was constant within a cultivar. The small variability of harvest index within a cultivar under these various conditions implied close coupling of the processes of vegetative and reproductive growth.

The stability of harvest index among plants within a cultivar from different environments in these experiments is in agreement with the results of Wilcox (1974), Buzzell and Buttery (1977), and Johnson and Major (1979). Across a range of plant densities from 25 to 582 thousand plants/ha, Wilcox (1974) found a change in harvest index of less than 0.05 for three cultivars. While the change was statistically significant, it was not of practical significance. Buzzell and Buttery (1977) showed that differences in the number of plants per hill from one to nine did not influence the harvest index. Johnson and Major (1979) reported that total biomass and seed yield of a genotype were influenced by planting date, but harvest index stayed the same. Only when frost prematurely killed the crop was harvest index reduced. The low threshold required to modify the harvest index of individual plants found in this study suggested that the density required to decrease the harvest index must be extremely high. This stability of harvest index among individuals of substantially different sizes was probably the basis for stability of harvest index across the broad range of planting densities found by Wilcox (1974).

Studies of other grain legumes which have been exposed to various degrees and timing of water stress

Table 3. Harvest index and yield of Wilkin plants which were stressed at different times during plant development.

Treatment	Stress period	Yield, g/m ²	Harvest index
0	None	205 (18)†	0.61 (0.006)†
1	Vegetative	164 (13)	0.62 (0.006)
2	Flowering	77 (9)	0.60 (0.006)
3	Seed set	161 (21)	0.61 (0.007)
4	Seed fill	108 (8)	0.56 (0.014)
5‡	None	—	0.63 (0.008)

† Standard error.

‡ Second sowing.

Table 4. Biomass reductions and harvest index stability of three cultivars for three sowing dates and four distances from the line source irrigation.

Cultivar	Sowing date	Biomass reduction† (kg ha ⁻¹ m ⁻¹)	Harvest index (SE)			
			Distance (m)			
			13	20	27	33
Chippewa	3/16	-78**	0.62 (0.002)	0.61 (0.003)	0.61 (0.009)	0.57 (0.003)
	4/2	-62**	0.59 (0.002)	0.57 (0.002)	0.57 (0.009)	0.56 (0.008)
	4/16	-52**	0.61 (0.008)	0.61 (0.003)	0.59 (0.010)	0.58 (0.010)
Evans	3/16	-42**	0.64 (0.009)	0.62 (0.007)	0.63 (0.002)	0.60 (0.013)
	4/2	-17*	0.61 (0.004)	0.68 (0.005)	0.60 (0.005)	0.59 (0.012)
	4/16	-93**	0.62 (0.012)	0.65 (0.002)	0.65 (0.002)	0.62 (0.004)
Wilkin	3/16	-60**	0.64 (0.005)	0.62 (0.002)	0.60 (0.000)	0.61 (0.002)
	4/2	-9	0.59 (0.003)	0.58 (0.003)	0.57 (0.005)	0.58 (0.018)
	4/16	-46**	0.59 (0.001)	0.59 (0.025)	0.61 (0.014)	0.61 (0.009)

† Biomass reductions were linear regression slopes of biomass on distance from the line source irrigation. Statistical significance for regression slopes are shown at $p < 0.01$ (**) and $p < 0.05$ (*).

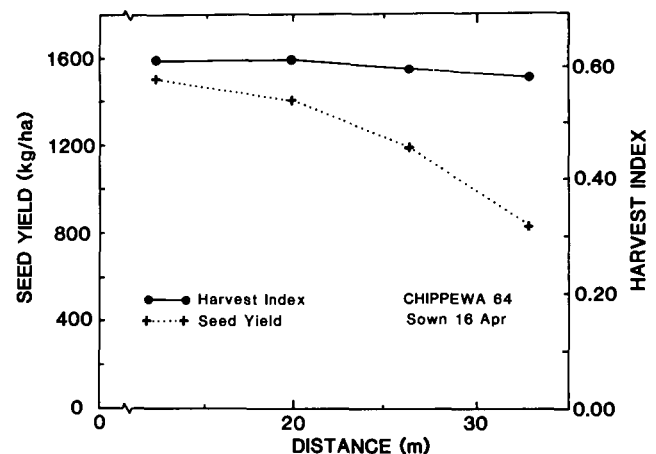


Fig. 3. Seed yield and harvest index of Chippewa 64 plants from the third sowing at four distances from a line source irrigation system.

corroborates the stability of harvest index. Turk and Hall (1980) irrigated cowpeas [*Vigna unguiculata* (L.) Walf.] during either vegetative growth, or flowering, or both with different quantities of water (line source). They found that regressions of seed mass on shoot mass accounted for 80% of the variation for all irrigation schedules. Such a result implies that harvest index was reasonably stable. Similarly, Harvey (1980) found no change in harvest index in peas [*Pisum sativum* (L.)] subjected to water stress in controlled environment chambers.

However, harvest index is not always independent of environment. Schapaugh and Wilcox (1980) found in soybeans that harvest index of late maturing cultivars dropped significantly in a poor year. Sinclair et al. (1981) have shown [using data from Schapaugh and Wilcox (1980)] that stability of harvest index was associated with relatively good performance in a poor year. They suggested that selection of improved lines across environments may indirectly select for stability in harvest index. The indeterminate growth pattern may provide greater stability of the harvest index because plants exhibiting this characteristic have the opportunity to adjust the number of seeds which are initiated over a longer period of time than do the determinate cultivars.

The stability of harvest index among plants of a population irrespective of plant sizes has been used implicitly in previous studies of harvest index. That is, researchers have used fewer plants in determination of harvest index than they have done for determinations of seed yield (e.g., Wilcox, 1974; Wallace and Munger, 1966). These data suggested that much broader application of harvest index can be made to simplify sampling and analysis of field plots by allowing smaller sample sizes. This will be especially useful in cases where experimental treatment or analysis might be difficult for large numbers of plants. While harvest index within a cultivar can change in response to environmental fluctuation or treatment, its general stability

across this broad range of environmental conditions also may prove to be useful in efforts to predict seed yield.

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